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Group-16

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SINGLE ANSWER TYPE

1. Ozone reacts with dry iodine to give?
(A) IO_2 (B) I_2O_3 (C) I_2O_4 (D) I_4O_9
2. Which of the following Isotopes is present in largest amount?
(A) O^{16} (B) O^{17} (C) O^{18} (D) O^{15}
3. Which does not liberate O_2 heating?
(A) MgO (B) NaNO_3 (C) Pb_3O_4 (D) KClO_3
4. Hydrogen sulphide reacts with lead acetate forming a black compound which reacts with H_2O_2 to form another compound. The colour of the new compound is
(A) Black (B) Yellow (C) White (D) Pink
5. One gas bleaches the colour of the flowers by reduction while the other by oxidation. The gases are
(A) CO and CO_2 (B) H_2S and Br_2 (C) SO_2 and O_3 (D) NH_3 and SO_3
6. Most acidic oxide is
(A) SeO_2 (B) SeO_3 (C) SO_2 (D) SO_3
7. Hypo is used in photography to
(A) Reduce AgBr grains to metallic Ag
(B) Convert metallic Ag to silver salt
(C) Remove undecomposed AgBr as soluble complex
(D) Remove reduced Ag
8. Which oxide is more acidic?
(A) Al_2O_3 (B) Na_2O (C) MgO (D) CaO
9. The compound of sulphur that can be used as refrigerant is
(A) S_2Cl_2 (B) SO_2 (C) SO_3 (D) H_2SO_4
10. Neutral litmus paper moistened with KI solution turns blue in ozone because of
(A) Iodine liberation (B) Oxygen liberation
(C) Alkali formation (D) Ozone reacts with litmus paper
11. The oxide insoluble in water is
(A) TeO_2 (B) SO_2 (C) PoO_2 (D) SeO_2
12. When H_2SO_4 added to charcoal
(A) There is no reaction (B) Water gas is formed
(C) SO_2 and CO_2 are evolved (D) CO and SO_2 are evolved.
13. Mustard gas is produced when SCl_2 reacts with ethene. It is
(A) $\text{S}(\text{CH}_2\text{CH}_2\text{Cl})_2$ (B) $(\text{CH}_3\text{CH}_2)_2\text{S}$ (C) CS_2 (D) H_2S
14. Which is used in vulcanisation of rubber
(A) SF_6 (B) SF_4 (C) SF_2 (D) S_2Cl_2

15. The substance which is blue in all three states is
(A) O_2 (B) O_3 (C) S_2 (D) Cl_2
16. The percentage of ozone in ozonized Oxygen is about
(A) 10 % (B) 40 % (C) 80 % (D) 100 %
17. Sulphuric acid on heating with sulphur powder gives
(A) H_2SO_3 (B) SO_3 (C) SO_2 and SO_3 (D) SO_2
18. Oxygen is gas but sulphur is solid because
(A) oxygen is composed of discrete molecules while sulphur is polymeric
(B) Molecular weight of sulphur is much higher than that of oxygen
(C) Oxygen is a stronger oxidising agent than sulphur
(D) Boiling point of sulphur is much higher than that of oxygen
19. Which substance chars when warmed with conc. H_2SO_4 ?
(A) Protein (B) Fat (C) Hydrocarbon (D) Carbohydrate
20. A certain compound when burnt gave three oxides. The first turned lime water milky the second, turned cobalt chloride paper pink and the third formed an aqueous solution of pH 3 nearly. The elements present in the compound are
(A) C, S, O (B) C, H, Na (C) C, H, S (D) C, H, Ca
21. The element having least tendency to exist in puckered 8-membered ring structure is
(A) S (B) Se (C) Po (D) Te

MULTIPLE ANSWER TYPE

22. Paramagnetic species among the following is/are
(A) O_2 (B) O_3 (C) S_6 (D) S_2
23. $TeCl_4$ is expected to be
(A) tetrahedral (B) trigonal bipyramidal
(C) square planar (D) octahedral
24. Which of the following give oxygen on heating?
(A) mercuric oxide (B) potassium chlorate
(C) potassium permanganate (D) ammonium dichromate
25. The following oxides do not exist as monomers in all states
(A) SO_3 (B) SO_2 (C) SeO_2 (D) SeO_3
26. Best ligand among the following is
(A) H_2Se (B) H_2Te (C) H_2S (D) H_2O
27. Which of the following is/are not hydrolyzed?
(A) SF_6 (B) Cl_2O (C) F_2O (D) Br_2O
28. The following is/are Lewis acid/s
(A) SO_2 (B) SO_3 (C) SF_6 (D) H_2SO_4
29. F_2 on reaction with water gives
(A) O_2 (B) O_3 (C) H_2 (D) HF
30. Which elements are polymorphic?
(A) O (B) S (C) Te (D) Se
31. Which of the following give coloured gases/fumes on reaction with conc. H_2SO_4 ?

- (D) N_2O_5

KEY

[illegible]